

Nathan James Anderson

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WORK EXPERIENCE

The Church of Jesus Christ of Latter-day Saints

NLP Product Manager

Provo, UT (Remote)

October 2022 – *current*

- Architected a 3-stage transfer learning pipeline (Whisper, SpaCy, mmBERT + LoRA) to assess oral language proficiency across 9 target languages and replace a legacy vendor system, cutting operational costs by 75% (\$250k+/year) while raising agreement with trained human raters from 0.36 to 0.86 (Cohen's kappa).
- Engineered pedagogically informed GEC feedback workflows using LLM prompt engineering, few-shot prompting, structured output parsing (OpenAI API), and human-in-the-loop quantitative evaluations, scaling deployment to 41 target languages for 5,000+ annual users.
- Built parallelized text mining and safety classification pipelines using spaCy and mmBERT for a multi-million-sentence corpora across 70+ languages.
- Directed applied NLP research operations, leading up to 8 Research Assistants in 2-week Agile sprints using Jira. Conducted biweekly literature reviews on relevant cutting-edge research.

Lowe's

Data Science Graduate Intern

Mooresville, NC (Remote)

June 2021 – August 2021

- Designed and trained a Siamese Neural Network using TensorFlow and K-Means Clustering to evaluate pairwise attribute similarity across 3,000+ catalog product features and optimize complementary product recommendation algorithms based on consumer purchase data. Model was adopted internally for testing.

The Church of Jesus Christ of Latter-day Saints

NLP Research Intern

Riverton, UT

July 2019 – December 2019

- Developed an English-to-38-language Speech-to-Speech Translation application, integrating custom Automatic Speech Recognition (ASR), Machine Translation (MT), and Text-to-Speech (TTS) models.
- Curated and aligned 50+ hours of audio/video/text data to fine-tune Azure Speech-to-Text models, reducing Word Error Rate (WER) from 7.2% to 3.3%.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java, Javascript, Bash/CLI

ML & NLP Tools: PyTorch, Hugging Face, SpaCy, Stanza, scikit-learn, XGBoost, OpenAI API, pandas, numpy

Speech: Whisper/WhisperX, Azure Speech, pyannote.audio, wav2vec2, jiwer

Cloud Platforms: Google Cloud Platform, Microsoft Azure, AWS

EDUCATION

Carnegie Mellon University MS, Language Technologies

GPA: 3.90

Pittsburgh, PA

August 2022

Brigham Young University BA, Linguistics, *cum laude*

GPA: 3.93

Provo, UT

April 2020

Minors: Computer Science, Linguistic Computing

Awards: Full-Tuition Scholarship, 2 FLAS Fellowships (Indonesian, Portuguese), Boren Scholarship (Turkish)